

Dynamics from Probability: Predictive Operators and the Koopman Picture

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Abstract

Given a probability measure on the observable σ -algebra — the output of Paper I — this paper derives the dynamics of the system. The central object is the *predictive kernel*: the regular conditional distribution of a future observable $F : \Omega \rightarrow \beta$ given a present observable $Q : \Omega \rightarrow \alpha$. From the predictive kernel we construct the *minimal predictive state map* Q_* , a canonical measurable map from the sample space into the space of probability measures on outcomes. The predictive factorization theorem shows that Q_* carries all conditional information about F : conditional expectations of functions of F factor through Q_* . Indexing over a discrete time horizon then yields a family of Markov kernels $\{\Pi_t\}$ and a corresponding family of *predictive operators* $\{K_t\}$. The Chapman–Kolmogorov equation for $\{\Pi_t\}$ is derived from temporal coherence of prediction, not assumed; it produces the semigroup law $K_{t+s} = K_t \circ K_s$. The dual *Koopman–Perron* relation $\int K_t g d\mu = \int g d(P_t^* \mu)$ connects operator-on-functions and measure-on-states into a single picture. When the kernels are Dirac measures, the operators reduce to classical Koopman operators and the semigroup law becomes the flow law $\varphi_{t+s} = \varphi_t \circ \varphi_s$. All results are formalized in Lean 4 / Mathlib with zero sorry obligations.

1 Introduction

The problem. Paper I produced a probability measure P on the observable σ -algebra from coherent structured observations. The question Paper II addresses is: what does P say about how the system *moves*? The answer is not an additional assumption. Once P is fixed, the temporal structure of prediction is already determined.

The Koopman picture. The classical approach to dynamics centers on the *Koopman operator* $U_T f = f \circ T$, which acts on observable functions by precomposing with the dynamics T . This paper takes the dual view: dynamics is what the observer *predicts* about the future, not what the system *does* in state space. The predictive operator K_t takes a function g of future outcomes and returns the observer’s expected value of g at each current predictive state. When the dynamics is deterministic, $K_t g = g \circ \varphi_t$ and the two pictures coincide. In the stochastic case, K_t and its dual P_t^* (acting on distributions) are genuinely adjoint: neither can be reduced to the other, and together they constitute the complete observable picture of the dynamics.

What this paper shows. Starting from a probability space $(\Omega, \mathcal{F}, P)$, a measurable map $Q : \Omega \rightarrow \alpha$ (the present observation), and a measurable map $F : \Omega \rightarrow \beta$ (the future observation), we construct:

- (i) the *predictive kernel* Π_Q , the regular conditional distribution of F given Q (Definition 3.1);
- (ii) the *minimal predictive state map* $Q_* = \varphi_Q \circ Q$, which encodes the conditional law as a single measurable map into $\mathcal{P}(\beta)$ (Definition 3.3);
- (iii) a family of predictive operators $\{K_t\}$ satisfying the semigroup law $K_{t+s} = K_t \circ K_s$ (Theorem 5.5);

- (iv) the Koopman–Perron duality between K_t and the push-forward P_t^* on distributions (Theorem 6.2);
- (v) the deterministic specialization: Dirac kernels recover classical Koopman operators (Theorem 7.1).

Relation to Paper I. Paper I took the output of coherent structured observations as input and produced a probability measure P on the observable σ -algebra. Paper II takes that measure as given and asks what temporal structure it carries. The setup is two measurable maps on the resulting probability space: $Q : \Omega \rightarrow \alpha$ encoding the present state of observation, and $F : \Omega \rightarrow \beta$ encoding the future. The formalism is self-contained.

Lean formalization. All results in this paper are formalized in Lean 4 / Mathlib in two files:

- `PredictiveState.lean` — the predictive kernel, minimal predictive state map, and predictive factorization theorem (zero sorry obligations).
- `PredictiveOperators.lean` — the semigroup, Koopman–Perron duality, and deterministic specialization (zero sorry obligations).

The key Mathlib ingredient is `ProbabilityTheory.Kernel.Disintegration`, which provides the Rokhlin disintegration theorem for standard Borel spaces, and `Probability.Kernel.Composition.Integration` which gives the Bochner–Fubini theorem for kernel compositions.

2 Setup

Throughout this paper, $(\Omega, \mathcal{F}, P)$ is a probability space. We fix two measurable maps:

- $Q : \Omega \rightarrow \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \rightarrow \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β , equipped with its canonical (Giry) σ -algebra. The *joint pushforward* is the measure $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$; its marginals are $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

3 The predictive kernel and the minimal predictive state map

3.1 The predictive kernel

Definition 3.1 (Predictive kernel). The *predictive kernel* $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ is the regular conditional distribution of F given Q :

$$\Pi_Q(q, A) = P(F \in A \mid Q = q),$$

defined as the disintegration kernel $\Pi_Q(q, \cdot) = \rho(\cdot \mid \alpha)$ of the joint pushforward $\rho = P \circ (Q, F)^{-1}$ over its first marginal.

Concretely: Π_Q is the unique (up to $P \circ Q^{-1}$ -null sets) Markov kernel $\alpha \rightarrow \mathcal{P}(\beta)$ such that

$$\int_A \Pi_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Proposition 3.2 (Predictive compatibility, P2.2). *Let $\pi : \alpha \rightarrow \alpha'$ be measurable with $Q' = \pi \circ Q$. Then for $(P \circ Q^{-1})$ -almost every $q \in \alpha$:*

$$\Pi_Q(q, \cdot) = \Pi_{Q'}(\pi(q), \cdot).$$

Proof. The joint pushforward $\rho' = P \circ (Q', F)^{-1}$ satisfies $\rho' = \rho \circ (\pi \times \text{id})^{-1}$. The claim follows from the $P \circ Q^{-1}$ -a.e. uniqueness of the disintegration of ρ' over its first marginal, combined with the covariance of `condKernel` under measurable maps (`eq_condKernel_of_measure_eq_compProd` in Mathlib). $\square$

3.2 The predictive law map and the minimal predictive state map

The predictive kernel Π_Q maps each value $q \in \alpha$ to a probability measure on outcomes. Packaging this as a single measurable map gives:

Definition 3.3 (Predictive law map and minimal predictive state map). The *predictive law map* is

$$\varphi_Q : \alpha \rightarrow \mathcal{P}(\beta), \quad \varphi_Q(q) = \Pi_Q(q, \cdot).$$

The *minimal predictive state map* is the composition

$$Q_* := \varphi_Q \circ Q : \Omega \rightarrow \mathcal{P}(\beta).$$

The value $Q_*(\omega)$ is the full conditional law $P(F \in \cdot \mid Q = Q(\omega))$.

Proposition 3.4 (Measurability, P2.1/P2.3). *The maps φ_Q and Q_* are measurable.*

Proof. φ_Q is the underlying map of the Markov kernel Π_Q , which is measurable by definition of a kernel (`Kernel.measurable` in Mathlib). Since $\mathcal{P}(\beta)$ carries the subspace σ -algebra from the Giry space of all measures on β , the wrap as a `ProbabilityMeasure` is measurable by `Measurable.subtype_mk`. Measurability of Q_* then follows from composition. $\square$

The minimal predictive state map identifies points with the same predictive law:

Proposition 3.5 (Predictive equivalence, P2.4). *For $\omega, \omega' \in \Omega$:*

$$Q_*(\omega) = Q_*(\omega') \iff \Pi_Q(Q(\omega), \cdot) = \Pi_Q(Q(\omega'), \cdot).$$

No quotient construction is needed: the state space is the image of φ_Q inside the standard Borel space $\mathcal{P}(\beta)$.

4 Predictive factorization

The predictive factorization theorem is the central result of this paper. It says that the conditional expectation of any bounded measurable function of F , given Q , factors through the minimal predictive state map Q_* .

Theorem 4.1 (Predictive factorization, P2.5). *For every bounded measurable $g : \beta \rightarrow \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] = \left(\omega \mapsto \int g dQ_*(\omega) \right) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g dQ_*(\omega) = \int g d\Pi_Q(Q(\omega), \cdot)$. We verify the two defining properties of the conditional expectation.

Measurability. The map $q \mapsto \int g d\Pi_Q(q, \cdot)$ is strongly measurable (Bochner integrability of g against a Markov kernel; `StronglyMeasurable.integral_kernel` in Mathlib). Since $Q_* = \varphi_Q \circ Q$, the map $h = (\text{predictive operator applied to } g) \circ Q_*$ is $\sigma(Q)$ -measurable.

Set-integral identity. For every $\sigma(Q)$ -measurable set $S = Q^{-1}(A)$:

$$\begin{aligned}
\int_S h \, dP &= \int_{Q^{-1}(A)} \int g \, d\Pi_Q(Q(\omega), \cdot) \, dP(\omega) \\
&= \int_A \int g \, d\Pi_Q(q, \cdot) \, d(P \circ Q^{-1})(q) \quad (\text{push-forward}) \\
&= \int_A \int_{A \times \beta} g(f) \, d\rho(q, f) \quad (\text{disintegration identity}) \\
&= \int_{Q^{-1}(A)} g(F(\omega)) \, dP(\omega) = \int_S g(F) \, dP.
\end{aligned}$$

The key step is the disintegration identity for $\rho = P \circ (Q, F)^{-1}$, which in Mathlib is `Measure.setIntegral_conditional`.

By the a.e. uniqueness of conditional expectation (`ae_eq_condExp_of_forall_setIntegral_eq`), $h = E[g(F) \mid \sigma(Q)]$ P -a.e. $\square$

Corollary 4.2 (Predictive sufficiency, P2.5). $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P -a.e.

Proof. Since $Q_* = \varphi_Q \circ Q$, we have $\sigma(Q_*) \subseteq \sigma(Q)$. The conditional expectation given $\sigma(Q)$ already factors through Q_* by Theorem 4.1, so it is $\sigma(Q_*)$ -measurable. $\square$

The canonical predictive operator K_{Q_*} is the map

$$(K_{Q_*}g)(q_*) = \int g \, dq_*, \quad q_* \in \mathcal{P}(\beta).$$

Theorem 4.1 then reads: $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P -a.e. This is the canonical operator associated with the predictive state.

Proposition 4.3 (Properties of K_{Q_*} , P2.6). *The canonical predictive operator K_{Q_*} is:*

- (i) positive: $g \geq 0 \Rightarrow K_{Q_*}g \geq 0$;
- (ii) unital: $K_{Q_*}\mathbf{1} = \mathbf{1}$;
- (iii) linear: $K_{Q_*}(g + h) = K_{Q_*}g + K_{Q_*}h$, $K_{Q_*}(\lambda g) = \lambda K_{Q_*}g$;
- (iv) contractive: $|K_{Q_*}g|(q_*) \leq \|g\|_\infty$.

5 The predictive semigroup

So far, the setup has been static: a single pair of measurable maps (Q, F) representing present and future. To introduce dynamics, we index over a discrete time horizon. For each $t \in \mathbb{N}$, let $F_t : \Omega \rightarrow \gamma$ be the observable at time t , and let Π_t be the predictive kernel of F_t given the current predictive state.

5.1 Predictive kernel families

Definition 5.1 (Predictive kernel family and semigroup kernel). A *predictive kernel family* on a state space γ is a family $\{\Pi_t\}_{t \in \mathbb{N}}$ of Markov kernels $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$ with $\Pi_0 = \delta$ (identity: $\Pi_0(q, \cdot) = \delta_q$).

A *predictive semigroup kernel* is a predictive kernel family satisfying the Chapman–Kolmogorov equation:

$$\Pi_{t+s}(q, \cdot) = \int_\gamma \Pi_t(q', \cdot) \, d\Pi_s(q, q') \quad \text{for all } q \in \gamma, \, t, s \in \mathbb{N}.$$

In kernel notation: $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$ where $\circ_\kappa$ denotes kernel composition.

Remark 5.2 (Derivation, not assumption). The Chapman–Kolmogorov equation is *derived* from the temporal coherence of prediction (the same observable compatibility that determined P in Paper I), not assumed. For a system evolving under dynamics T , the kernel at time $t + s$ sends state q to the distribution of q after t and then s steps of prediction, which is exactly the composition $\Pi_t \circ_\kappa \Pi_s$.

5.2 The predictive operator family

Definition 5.3 (Predictive operator family). Given a predictive kernel family $\{\Pi_t\}$, the *horizon-indexed predictive operator* at horizon t is

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Proposition 5.4 (Basic properties, P3.1/P3.4). *For any predictive kernel family $\{\Pi_t\}$:*

- (i) Identity at zero: $K_0 g = g$ for all measurable g ;
- (ii) Positive: $g \geq 0 \Rightarrow K_t g \geq 0$;
- (iii) Unital: $K_t \mathbf{1} = \mathbf{1}$;
- (iv) Contractive: $\|K_t g\|_\infty \leq \|g\|_\infty$.

Proof. Identity at zero: $K_0 g(q) = \int g d\delta_q = g(q)$. Positivity and unitality are immediate from the Markov property and non-negativity of integration against a positive measure. Contractiveness: $|(K_t g)(q)| = |\int g d\Pi_t(q, \cdot)| \leq \|g\|_\infty \cdot \Pi_t(q, \gamma) = \|g\|_\infty$. $\square$

Theorem 5.5 (Semigroup property, P3.3). *Let $\{\Pi_t\}$ be a predictive semigroup kernel. For every bounded measurable $g : \gamma \rightarrow \mathbb{R}$:*

$$K_{t+s} g = K_t(K_s g).$$

Proof. Using the Chapman–Kolmogorov equation $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$:

$$\begin{aligned} (K_{t+s} g)(q) &= \int g d\Pi_{t+s}(q, \cdot) \\ &= \int g d(\Pi_t \circ_\kappa \Pi_s)(q, \cdot) \\ &= \int \left(\int g d\Pi_t(q', \cdot) \right) d\Pi_s(q, q') \quad (\text{Bochner–Fubini for kernel composition}) \\ &= \int (K_t g)(q') d\Pi_s(q, q') = (K_t(K_s g))(q). \end{aligned}$$

The Bochner–Fubini step is `Kernel.integral_comp` from `Mathlib.Probability.Kernel.Composition.Integration`. The boundedness hypothesis on g is needed for integrability of g against the composed kernel. $\square$

Remark 5.6 (Boundedness hypothesis). The hypothesis that g is bounded (i.e., $\|g\|_\infty < \infty$) is used to guarantee that g is integrable against the composed Markov kernel, which has total mass 1. The result extends to $g \in L^\infty(\gamma)$ by a monotone class argument.

6 Koopman–Perron duality

The predictive operators K_t act on functions. The dual objects act on measures.

Definition 6.1 (Koopman–Perron dual). Given a predictive semigroup kernel $\{\Pi_t\}$ and a (finite) measure μ on γ , the *Koopman–Perron push-forward* is

$$P_t^* \mu := \mu \star \Pi_t = \int_{\gamma} \Pi_t(q, \cdot) d\mu(q).$$

In Mathlib notation: $\mu.\text{bind } (\text{ker } t)$.

Theorem 6.2 (Koopman–Perron duality, P3.5). For every bounded measurable $g : \gamma \rightarrow \mathbb{R}$ and finite measure μ on γ :

$$\int_{\gamma} (K_t g) d\mu = \int_{\gamma} g d(P_t^* \mu).$$

Proof.

$$\begin{aligned} \int g d(P_t^* \mu) &= \int g d(\mu \star \Pi_t) = \int g d\left(\int \Pi_t(q, \cdot) d\mu(q)\right) \\ &= \int_{\gamma} \int g d\Pi_t(q, \cdot) d\mu(q) \quad (\text{Bochner–Fubini}) \\ &= \int_{\gamma} (K_t g)(q) d\mu(q). \end{aligned}$$

The Bochner–Fubini step again uses `Kernel.integral_comp`. □

Remark 6.3 (The two sides of the duality). K_t acts on the *function algebra*: it takes a function g on the state space and returns its predicted value t steps hence. P_t^* acts on the *distribution*: it takes a current distribution of predictive states and evolves it t steps forward. Neither is primary; they are adjoint. In the classical (deterministic) specialization below, K_t becomes the Koopman operator U_T^t and P_t^* becomes the pushforward $T_*^t \mu$.

7 Deterministic specialization

When the dynamics is deterministic, the stochastic picture collapses to the classical Koopman picture.

Theorem 7.1 (Deterministic specialization, P3.6). Suppose $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \rightarrow \gamma$. Then for every measurable g :

$$K_t g = g \circ \varphi_t.$$

Proof. $(K_t g)(q) = \int g d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The integral against a Dirac measure is `integral_dirac` in Mathlib. □

Corollary 7.2 (Flow law, P3.6). If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for all t , then

$$\varphi_{t+s} = \varphi_t \circ \varphi_s.$$

Proof. From $\Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s$ and the Dirac hypothesis:

$$\begin{aligned} \delta_{\varphi_{t+s}(q)} &= \Pi_{t+s}(q, \cdot) = (\Pi_t \circ_{\kappa} \Pi_s)(q, \cdot) \\ &= \Pi_s(q, \cdot) \star \Pi_t = \delta_{\varphi_s(q)} \star \Pi_t = \Pi_t(\varphi_s(q), \cdot) = \delta_{\varphi_t(\varphi_s(q))}. \end{aligned}$$

Since Dirac measures at distinct points are distinct (under `SeparatesPoints`), we conclude $\varphi_{t+s}(q) = \varphi_t(\varphi_s(q))$. □

Remark 7.3 (Classical Koopman). In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$ where U_T is the classical Koopman operator. The semigroup law $K_{t+s} = K_t \circ K_s$ becomes $U_T^{t+s} = U_T^t \circ U_T^s$, and the Koopman–Perron duality $\int K_t g d\mu = \int g d(P_t^* \mu)$ recovers the change-of-variables formula $\int g \circ T^t d\mu = \int g d(T_*^t \mu)$.

8 The observable dynamical system

The results above assemble into a single structure.

Definition 8.1 (Observable dynamical system). An *observable dynamical system* is a triple $(O_{Q_*}, \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$ where:

- $O_{Q_*} = \mathcal{P}(\beta)$ is the *predictive state space* (the image of Q_*);
- $\nu_{Q_*} = P \circ Q_*^{-1}$ is the *predictive state distribution* (the pushforward of P along Q_*);
- $\{K_t\}$ is the *predictive operator family* on O_{Q_*} , satisfying the semigroup law and Markov properties above.

The predictive state distribution ν_{Q_*} is stationary under P_t^* : if (Ω, P) is stationary (i.e., $P \circ T^{-1} = P$), then $P_t^* \nu_{Q_*} = \nu_{Q_*}$ for all t . This connects the observer’s prediction to invariant measure theory; the details are deferred to the continuous-time setting.

9 Relation to Paper III

Paper II has constructed the dynamics entirely from the probability measure P and the measurable maps (Q, F) . The predictive state space O_{Q_*} is a subset of $\mathcal{P}(\beta)$, equipped with the semigroup $\{K_t\}$.

Paper III asks the converse question: *when does the sequence of time-delayed observations $\{Q \circ T^{-n} : n \in \mathbb{N}\}$ determine the full state space?* The answer — under a *cyclic vector condition* on the Koopman operator — is that the Stone space of the observable algebra generated by time-delayed queries is isomorphic (as a measure space) to the original state space.

The connection to Paper II is direct. The Koopman operator appearing in the cyclic vector condition is the operator $U_T f = f \circ T$ on $L^2(\Omega, P)$ induced by the dynamics T . The predictive operators K_t act on functions on the predictive state space $O_{Q_*} \subset \mathcal{P}(\beta)$; when Q_* is injective up to P -null sets, K_t and U_T^t determine the same operator up to the identification of O_{Q_*} with the L^2 orbit of Q . When the cyclic vector condition holds, no information about the state is lost in the observable layer, and O_{Q_*} is (up to measure-theoretic identification) the full state space.

Remark 9.1 (Continuous-time deferred). All results above are stated for discrete time $t \in \mathbb{N}$. The continuous-time ($t \in \mathbb{R}_{\geq 0}$) versions require the infinitesimal generator of a C_0 -semigroup and the Hille–Yosida theorem. These are deferred to a subsequent treatment; the discrete-time results proved here are the foundation.